

A New Bioinformatics Tool for Prediction with Confidence

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Abstract—*Complementing each prediction with the measure of its reliability is crucial for domains with high misclassification costs. Conformal predictors represent a new flexible bioinformatics tool that can output predictions with a single value of confidence assigned to each prediction. Conformal predictors can also guarantee the error rate of region predictions by setting the confidence level.*

In this work, we applied different types of conformal predictors to high dimensional veterinary and medical microarray data. Implemented predictors resulted in the accuracy as high as provided by conventional machine learning methods, but in addition, they complemented each prediction with the measure of confidence. Validity of these predictions was also empirically confirmed: the ratio of mistakes made by region predictors was close to the preset level.

Keywords: Conformal predictor, confidence machine, confidence, microarray.

1. Introduction

Most machine learning classifiers predict a label without indicating how reliable the prediction is, and this can be crucial for such important fields with high misclassification costs as medical and veterinary diagnosis. For this reason, we are aiming to complement each individual prediction with a measure of its reliability — confidence.

This problem can be solved by applying conformal predictors introduced in [1] — a new tool that produces predictions with confidence. The confidence we assign should not be confused with confidence in statistical conclusions: in statistics we produce confidence intervals instead of a single value that estimate a parameter, and this interval is the same for the whole statistical model, that is, for all predictions.

The output of conformal predictors can also be represented as a region predictor, that is, a classifier that produces a set of labels as a prediction. The advantage of conformal predictors is that such region predictors are always *valid*, that is, the error rate of predictors corresponds (up to statistical fluctuations) to the significance level that was set beforehand. The property of validity is proven under a simple statistical i.i.d. assumption: the examples (combinations of an object and a label) are taken independently from the same distribution.

Most of predicted sets of labels output by conformal predictors are usually singletons. However, when there is

not sufficient information provided for predicting one label without loss in validity, conformal predictors may output multiple (or empty) predictions, that is, predictions that comprise more than one label (or contain no labels). Experiments in [1] demonstrate that multiple predictions cease to occur when the number of considered samples exceeds a certain level. This way conformal predictors help determine the number of samples required to train the algorithm so that it produces predictions of high reliability.

Conformal predictors allow us to process high-dimensional data. Thus, they are able to analyze gene expression profiles obtained using microarray technique. For this reason, we applied different conformal predictors to high-dimensional veterinary and medical microarray data.

The first aim of our experiments was to verify that conformal predictors empirically prove to have the property of validity on microarray data and to understand how many samples are needed to provide reliable predictions.

Secondly, even though conformal predictors output region predictions, we can force conformal predictors to produce one prediction at a time. We observed accuracy of conformal predictors in this experimental setup and compared with conventional machine learning methods that always produce singletons as predictions.

The rest of the paper is structured as follows. In Section 2 we provide the outline of conformal prediction framework. Section 3 introduces the data. Section 4 contains the description of experiments and discussion of their results. Finally, conclusions are made in Section 5.

2. Outline of Conformal Prediction

Let us assume that we are given a training set of examples $(x_1, y_1), \dots, (x_{n-1}, y_{n-1})$ where x_i is a vector of attributes and y_i is a label out of a finite set of possible labels (classes), and our goal is to predict the classification y_n for remaining example x_n .

The general idea of conformal predictors is to try every possible label y as a candidate for x_n 's label and see how random the resulting example (x_n, y) is in comparison to $(x_1, y_1), \dots, (x_{n-1}, y_{n-1})$. The ideal case is when all y 's except for one lead to sequences that are not random — we can then be confident in this prediction.

A *nonconformity measure* is a set of measurable mappings $\{A_n : n \in N\}$ of the type

$$A_n : Z^{(n-1)} \times Z \rightarrow (-\infty, +\infty] \quad (1)$$

where $Z^{(n-1)}$ is the set of all bags (multisets) of elements of Z of size $n-1$. This nonconformity measure will assign some value α_i (*nonconformity scores*) to every example in the sequence $\{z_i := (x_i, y_i), i = 1, \dots, n\}$ including a new example and will evaluate ‘nonconformity’ between a set and its element:

$$\alpha_i := A_n(\{z_1, \dots, z_{i-1}, z_{i+1}, \dots, z_n\}, z_i), \quad (2)$$

$i = 1, \dots, n$, where $\{\dots\}$ denotes a multiset. A nonconformity measure can be based on many well-known machine learning algorithms.

When considering a hypothesis $y_n = y$ and after finding the corresponding nonconformity scores $\alpha_1, \dots, \alpha_n$ for a full sequence with label y for the last example, a natural way to compare α_n to the other α_i s is to look at the ratio of examples who are as least as strange as the new example, that is, to calculate

$$p_n = \frac{|\{i = 1, \dots, n : \alpha_i \geq \alpha_n\}|}{n}. \quad (3)$$

This ratio is called the p -value associated with the possible label y for x_n . Thus, we can compliment each label with a p -value that shows how random the example with this label is in comparison with all previous examples, that is, how well a new example with this label conforms with the rest of the sequence.

The *conformal predictor* determined by the nonconformity measure A_n , $n \in N$ and a significance level ϵ is then defined as a measurable function $\Gamma : Z^* \times X \times (0, 1) \rightarrow 2^Y$ (2^Y is a set of all subsets of Y) such that the prediction set $\Gamma^{(\epsilon)}(x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n)$ is defined as the set of all labels $y \in Y$ such that $p_n > \epsilon$. Thus, for any finite sequence of examples with labels, $(x_1, y_1, \dots, x_{n-1}, y_{n-1})$, a new object without a label x_n and a significance level ϵ , the conformal predictor outputs a region prediction $\Gamma^{(\epsilon)}$ — a set of possible labels for a new object.

Conformal predictors defined above are *valid*: for any exchangeable probability distribution P on Z^∞ (Z^∞ is the set of all infinite sequences of elements of Z) and any significance level ϵ ,

$$\limsup_{n \rightarrow \infty} \frac{\sum_{i=1}^n \text{err}_n^\epsilon(\Gamma)}{n} \leq \epsilon \quad (4)$$

with probability one, where $\text{err}_n^\epsilon(\Gamma)$ is equal to 1 when the prediction set Γ^ϵ does not contain a real label y_n , and 0 otherwise. To put it simply, in the long run the frequency of errors made by a conformal predictor (that is, cases when prediction set Γ^ϵ does not contain a real label) does not exceed ϵ . The only assumption used is the i.i.d. assumption: all the examples have been generated independently by some probability distribution.

The property of validity is proven only for *the on-line mode*, that is, when we observe each example one by one and make each prediction taking into account the information only regarding the examples considered before, rather than predicting on the basis of a certain rule extracted from a fixed set of examples. The latter mode is called *the off-line mode*. Nevertheless, validity is empirically proven to remain in the off-line mode [1].

For each object one can calculate confidence and credibility. *Credibility* is a maximum value of all possible p -values. *Confidence* equals $1 -$ the second maximum p -value. High confidence means that the alternative hypothesis is excluded by having a low p -value, high credibility checks whether the prediction itself does not have too small a p -value.

3. Data

Two high-dimensional microarray data sets were used: veterinary data of Salmonella subtypes and medical data of human cancers.

Salmonella data are provided by the Veterinary Laboratories Agency of the DEFRA. These are strains collected from epidemiological surveys from food animals that are likely to be a source of Salmonella contamination for humans. Each strain is represented by at least two replicates. There are 56 strains (120 replicates) in total. Each replicate has 3858 features — real numbers between 0 and 1. Only genes without missing information are used.

We consider the following Salmonella subtypes as classes: Enteritidis — 24 strains (51 replicates), Newport — 14 strains (33 replicates), Typhimurium — 18 strains (36 replicates). Features were normalized, averaged across replicates, and rounded to 0 and 1.

Human cancers data set was provided in the ICMLA 2009 Challenge [2]. These samples were profiled on the Affymetrix Human Genome GeneChip HG_U133 Plus 2.0, which contains 54,000 probes covering 20,000 annotated genes. All the samples were profiled by the International Genomics Consortium (IGC) for the Expression Project for Oncology (expO) and were deposited in the NCI GEO Omnibus under the series of GSE2109.

We analyzed the training set of 400 samples, comprising 70 lung cancers, 130 colon cancers, and 200 breast cancers. The aim was to predict the type of cancer. 447 features were preselected out of initial 54613 to reduce the processing time.

4. Experiments

4.1 Implemented Conformal Predictions

The conformal prediction framework is flexible, because we can use most of known machine learning algorithms as an underlying algorithm for designing a nonconformity measure (which is a core element in a conformal predictor).

We used several types of non-conformity measures in microarray data analysis:

- measures based on k NN algorithm [3]; we will denote them CP- k NN, where k is the number of nearest neighbors;
- based on SVM with different kernels [4], [5]; denoted as CP-SVM (*kernel*, *parameter*);
- based on random forests [6]; referred to as CP-RF;
- based on k NN applied to random forest proximities [7]; denoted as CP-RF- k NN, where k is the number of nearest neighbors).

The experiments were carried out in two settings: off-line in leave-one-out (LOO) procedure, to show the usage of conformal predictors as conventional classifiers, and on-line, to demonstrate the advantages of region predictions and to identify the sufficient number of examples to provide valid predictions on this data set.

4.2 Results

First, we will demonstrate how conformal predictors work.

Recall that conformal predictors provide a p -value for each object and each hypothetical label. The p -values calculated for the Salmonella data in the on-line mode are shown in Table 1. After assigning a p -value for each possible label, we can predict the label with the highest p -value. We will call a prediction extracted from a conformal predictor this way *forced point prediction*, because originally conformal predictors are applied to produce region predictions. If several labels have the highest p -value, we call this situation a *tie*.

For each prediction we can then calculate confidence and credibility as described in Section 2. Confidence shows how confident we are in rejecting the other label, and credibility demonstrates how well the chosen label conforms with the rest of the set. If confidence is close to 1 and credibility is not close to 0, the prediction can be considered as reliable.

Table 1 represents p -values, confidence and credibility for CP-RF applied to the first 25 examples of the Salmonella data in the online mode: the first column shows the number of examples we consider, the others contain p -values for three classes, confidence, credibility, predicted label (or a tie) and a true label.

We will first consider individual examples separately. Example 21 has three p -values one of which is high (95.2%), the others are low (4.8%). This results in high confidence and high credibility of 95.2% and identifies the prediction as reliable: only one label conforms well with the rest of the set. If this example were classified as class 1 or 2, this would mean that an event of probability $\leq 4.8\%$ occurred. For this reason, we expect the example to belong to class 3, which is true.

In contrast, both examples 12 and 22 have low p -values for each class, which means that we cannot be confident when assigning these examples to any of the classes. These

Table 1: Classification with confidence for the CP-RF applied to the Salmonella data in the on-line mode: each object is complemented with p -values, one for each label, as well as values of confidence and credibility

No.	p -value for 1	p -value for 2	p -value for 3	Confidence	Credibility	Predicted label	True label
3	100.0%	66.7%	100.0%	0.0%	100.0%	Tie	1
4	75.0%	75.0%	75.0%	25.0%	75.0%	Tie	2
5	40.0%	100.0%	40.0%	60.0%	100.0%	2	2
6	33.3%	50.0%	33.3%	66.7%	50.0%	2	1
7	28.6%	28.6%	14.3%	71.4%	28.6%	Tie	3
8	75.0%	12.5%	12.5%	87.5%	75.0%	1	1
9	11.1%	11.1%	11.1%	88.9%	11.1%	Tie	1
10	70.0%	10.0%	10.0%	90.0%	70.0%	1	1
11	9.1%	9.1%	45.5%	90.9%	45.5%	3	3
12	8.3%	8.3%	8.3%	91.7%	8.3%	Tie	2
13	7.7%	7.7%	84.6%	92.3%	84.6%	3	3
14	7.1%	7.1%	100.0%	92.9%	100.0%	3	3
15	6.7%	26.7%	6.7%	93.3%	26.7%	2	2
16	81.3%	6.3%	6.3%	93.8%	81.3%	1	1
17	5.9%	41.2%	5.9%	94.1%	41.2%	2	2
18	5.6%	5.6%	100.0%	94.4%	100.0%	3	3
19	5.3%	15.8%	5.3%	94.7%	15.8%	2	2
20	10.0%	5.0%	5.0%	95.0%	10.0%	1	1
21	4.8%	4.8%	95.2%	95.2%	95.2%	3	3
22	4.5%	4.5%	9.1%	95.5%	9.1%	3	3
23	30.4%	4.3%	4.3%	95.7%	30.4%	1	1
24	4.2%	4.2%	50.0%	95.8%	50.0%	3	3
25	4.0%	4.0%	68.0%	96.0%	68.0%	3	3

low p -values produce high confidence (91.7% and 95.5%) and low credibility (8.3% and 9.1%). For this reason, we are likely to make an erroneous prediction. As a result, the output for example 12 is a tie, but prediction for example 22 is correct.

In general, one can see from Table 1 that when the number of examples is relatively low (up to 12 examples), the situations occur when forced predictions make errors or output ties. Starting from example 13 we never output erroneous predictions, confidence is always high (it has the maximum possible value), but credibility varies a lot. High value of confidence implies that we are confident in rejecting the labels that are not true. This tendency can be observed starting from example 8.

Table 2: Accuracy of forced correct predictions for conformal predictors applied to Salmonella and human cancers data in the LOO mode.

Algorithm	Salmonella Data	Cancer Data
CP-RF	98.2–100.0%	91.3–91.3%
CP-RF-1NN	98.2–100.0%	92.5–93.0%
CP-RF-5NN	98.2–100.0%	91.3–91.5%
CP-1NN	98.2–100.0%	86.0–86.3%
CP-5NN	98.2–100.0%	89.3–89.5%
CP-SVM (rbf, 5)	92.9–94.6%	88.0–88.5%
CP-SVM (poly, 5)	98.2–100.0%	89.3–89.8%
Max bare prediction	100.0%	92.5%

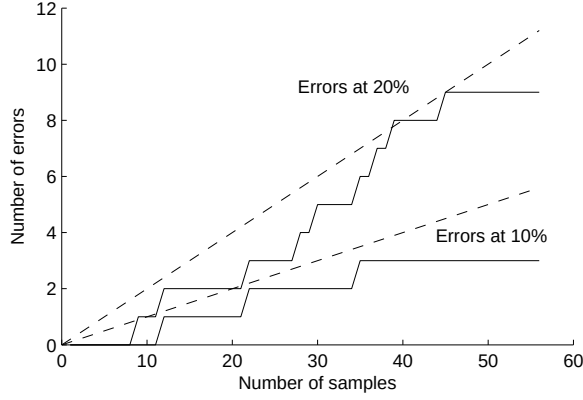


Figure 1: Validity of CP-RF-1NN applied to the Salmonella data in the on-line mode: the horizontal axis represents the number of examples in the on-line mode, dotted and solid lines demonstrate the expected and actual numbers of errors for different significance levels, respectively.

These experiments demonstrated that the Salmonella data set is clean and it is enough to process 12 samples to start making correct predictions with high confidence. This will be further confirmed when identifying the number of samples starting from which conformal predictors stop producing multiple predictions.

Since the Salmonella data set is so pure, when we launched experiments in the LOO mode most conformal predictors provided high accuracy of forced point predictions on Salmonella data: 55 correct predictions and 1 tie for 56 objects. Accuracy on cancer data varied from 88.0% to 93%. The accuracy of different conformal predictors can be seen in Table 2. The lower boundary of the range shown is the accuracy when all ties make erroneous predictions, the upper — when all ties are correct. The last row of the table represents the maximum accuracy of machine learning methods achieved on the data sets without conformal prediction framework. Among these methods we considered random forests, SVM and k NN. Comparison shows that conformal predictors are not beaten by conventional machine learning methods in accuracy when forced to output singletons as predictions. At the same time, they assign confidence to each prediction and produce the second type of predictions — region predictions, which are always valid.

To demonstrate the validity of conformal predictors, we launched them as region predictors that, for each significance level $\epsilon > 0$, predict a set of labels with p -values greater than ϵ . Thus, the predictor may output as a prediction no label (we call this an *empty prediction*), one label (*certain prediction*) or more than one label (*multiple prediction*).

The conformal predictors implemented in our work proved to be valid, that is, for a given significance level $\epsilon > 0$ the rate of erroneous predictions (predictions not containing an actual label) does not exceed ϵ . This was the case in all

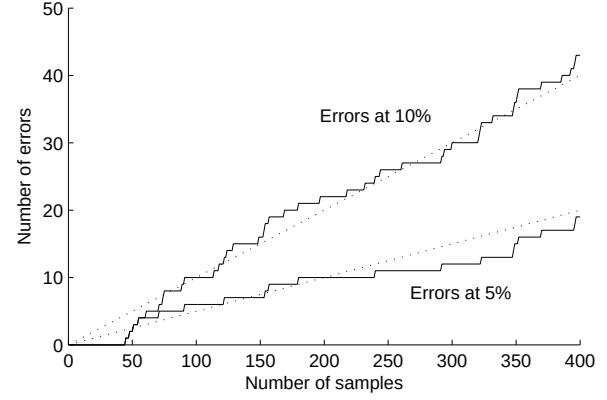


Figure 2: Validity of CP-RF-1NN applied to the human cancers data in the on-line mode: the horizontal axis represents the number of examples in the on-line mode, dotted and solid lines demonstrate the expected and actual numbers of errors for different significance levels, respectively.

experiments on all data sets.

The validity example is demonstrated in Figures 1 and 2. The figures show the erroneous predictions dynamics and confirms the validity of the CP-RF-1NN applied to the cancer and Salmonella data: solid lines, which represent the actual number of errors, are close to dotted lines, which demonstrate the expected number of errors.

Not only did our predictions have a guaranteed error rate, but also most predictions obtained in experiments with both Salmonella and human cancers were certain, that is, the algorithm usually output one label as a prediction. In

Table 3: Efficiency of conformal predictors in the LOO mode: certain prediction and certain correct prediction rates for different conformal predictors at significance levels $\epsilon = 5\%$ and 10% .

Significance level	10%		5%	
	Certain	Certain correct	Certain	Certain correct
Cancer data				
CP-RF	94.8%	89.8%	89.5%	86.8%
CP-RF-1NN	94.0%	89.3%	91.3%	87.8%
CP-RF-5NN	94.5%	89.5%	89.5%	86.0%
CP-1NN	86.5%	79.0%	68.8%	64.5%
CP-5NN	96.0%	87.8%	78.3%	74.5%
CP-SVM (rbf, 5)	89.5%	83.8%	77.0%	72.8%
CP-SVM (poly, 5)	81.5%	76.5%	67.5%	64.3%
Salmonella data				
CP-RF	91.1%	91.1%	96.4%	96.4%
CP-RF-1NN	91.1%	91.1%	98.2%	98.2%
CP-RF-5NN	92.9%	92.9%	98.2%	98.2%
CP-1NN	91.1%	91.1%	96.4%	96.4%
CP-5NN	91.1%	91.1%	96.4%	96.4%
CP-SVM (rbf, 5)	94.6%	91.1%	94.6%	91.1%
CP-SVM (poly, 5)	91.1%	91.1%	96.4%	96.4%

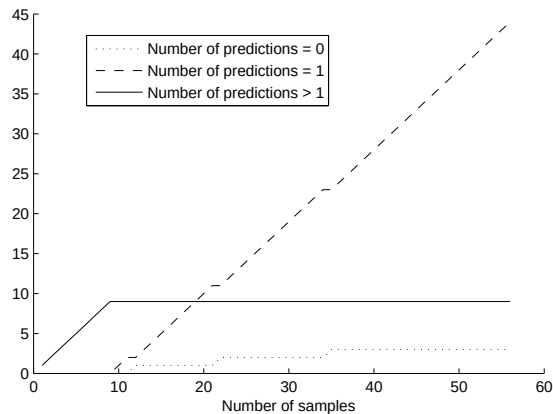


Figure 3: Efficiency at significance level 10% for the CP-RF-1NN applied to the Salmonella data in the on-line mode.

addition, almost all certain predictions were correct, which implies that most errors were made by empty predictions. The exact rates of certain predictions and correct certain predictions are shown in Table 3.

Figures 3 and 4 demonstrate the dynamics of efficiency characteristics at significance level of 10% of the CP-RF-1NN applied in the on-line mode to the Salmonella and cancer data, respectively. The characteristics shown are the number of multiple predictions, the number of certain predictions and the number of empty predictions. The figures demonstrate that while the number of analyzed examples is low (up to 9 examples for Salmonella data), they do not carry enough information to make certain predictions without losing validity, that’s why most predictions are multiple. But starting from Salmonella example 10, we have accumulated enough information so that multiple predictions became rare and most of prediction regions contain exactly one label. The Salmonella plot confirms purity of the Salmonella data: the first ten examples contain enough information to make valid correct predictions at the level 10%; no multiple predictions are output after example 9 and all certain predictions are correct. The cancer data set is not as clean: some of certain predictions are not correct, and multiple predictions do not stop occurring, but are output extremely rarely.

Both plots conform with the empirical fact established in [1] that empty predictions start to occur when the number of multiple predictions stabilizes. In addition, the figures allow us to determine the number of examples required for producing mostly correct predictions in the on-line mode: 10 examples for Salmonella data and about 50 examples for cancer data.

5. Conclusion

When providing predictions, we have to balance validity and efficiency: the bigger the prediction set, the fewer erroneous predictions we make. Conventional machine learning

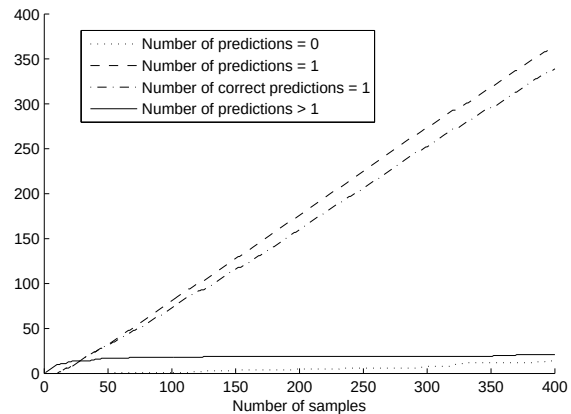


Figure 4: Efficiency at significance level 10% for the CP-RF-1NN applied to the cancer data in the on-line mode.

methods always output one label as a prediction and, hence, have fixed efficiency. Conformal predictors, in contrast, can guarantee validity.

However, conformal predictors can also be forced to output a singleton as a prediction with a loss in validity guarantee. Thus, the output of conformal predictors can be interpreted in two different ways: either with guaranteed validity, or with guaranteed efficiency.

Experiments on medical and veterinary microarray data demonstrated that conformal predictors can be a useful bioinformatics tool. When producing region predictions, the property of validity was empirically confirmed in all experiments: the ratio of mistakes made by region predictors does not exceed the preset level. When forced to output predictions-singletons, the performance of conformal predictors was comparable to the performance of conventional machine learning methods, which resulted in high accuracy of forced point predictions.

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